

Anti-Spoofing Face Attendance using CNN

A Mini project report submitted in partial fulfilment of the requirement for the
award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING (AI&ML)

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CERTIFICATE

This is to certify that the B.Tech Mini Project report entitled “**Anti-Spoofing Face Attendance using CNN**” is a Bonafide work done by **M. MANIDEEP REDDY (23U11A6663), G. KRISHNAVENI (23U11A6632), M. NANDINI (23U11A6657), J. BALARAMA KRISHNA (23U11A6642), N. VARUN KUMAR (22U11A6631)** in partial fulfilment of the requirement of the award for the degree of Bachelor of Technology in “**Computer Science and Engineering (AI&ML)**” from Jawaharlal Nehru Technological University, Hyderabad during the year 2025-2026

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This is a record of Bonafide work carried out by us and the results embodied in this project have not been reproduced or copied from any source. The results embodied in this project report have not been submitted to any other University or Institute for the award of any other degree or diploma.

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ABSTRACT

Face Attendance Systems have become increasingly popular for automating attendance management in educational institutions and organizations. However, traditional face recognition systems are vulnerable to spoofing attacks such as printed photographs, mobile screen displays, and replay videos, which can result in unauthorized attendance marking. To address this issue, an Anti-Spoofing Face Attendance System using Convolutional Neural Networks (CNN) is proposed.

The developed system integrates face detection, anti-spoofing verification, face recognition, and attendance management into a single platform. Facial images are captured through a webcam and processed using image preprocessing techniques. A CNN model is trained to classify facial inputs as genuine or spoofed. Only genuine faces are forwarded to the face recognition module for user identification. Upon successful authentication, attendance records are automatically stored in the MySQL database.

The system is implemented using Python, OpenCV, TensorFlow, Keras, MySQL, and related machine learning libraries. Experimental results demonstrate that the proposed system effectively detects spoofing attempts while accurately recognizing registered users and recording attendance. The integration of anti-spoofing mechanisms significantly improves the security and reliability of attendance management systems.

Keywords: Anti-Spoofing, Face Attendance System, Face Recognition, Convolutional Neural Network (CNN), Deep Learning, Machine Learning, OpenCV, TensorFlow, Keras, Image Processing, Facial Authentication, Face Detection, Attendance Management System, Biometric Authentication, Liveness Detection, Real-Time Recognition, Feature Extraction, MySQL Database, Computer Vision, Security, Spoof Attack Detection.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
CNN	Convolutional Neural Network
CVPR	Conference on Computer Vision and Pattern Recognition
DFD	Data Flow Diagram
ER	Entity Relationship
GUI	Graphical User Interface
IEEE	Institute of Electrical and Electronics Engineers
ISO	International Organization for Standardization
ML	Machine Learning
MySQL	My Structured Query Language
SDLC	Software Development Life Cycle
SRS	Software Requirement Specification
UML	Unified Modeling Language

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CHAPTER 1

INTRODUCTION

1.1 Overview of the Project

Attendance management is an essential activity in educational institutions, corporate organizations, and training centres. Accurate attendance records help organizations monitor participation, maintain discipline, and evaluate performance. Traditional attendance systems such as manual registers, RFID cards, and biometric devices often suffer from limitations including time consumption, human errors, proxy attendance, and maintenance costs.

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), Deep Learning, and Computer Vision have enabled the development of intelligent attendance systems based on facial recognition technology. Face recognition provides a contactless and automated method for identifying individuals using their unique facial features. However, conventional face recognition systems are vulnerable to spoofing attacks where attackers attempt to deceive the system using printed photographs, mobile screens, videos, or digital displays.

To address these challenges, the proposed Anti-Spoofing Face Attendance System integrates facial recognition with Convolutional Neural Network (CNN) based liveness detection. The system verifies whether the detected face is genuine before performing recognition and attendance recording. This additional security layer significantly reduces the possibility of proxy attendance and unauthorized access.

The system captures live video through a webcam, performs face detection, validates liveness using a CNN model, recognizes registered users, and records attendance automatically. The project also incorporates session-based attendance management, late attendance detection, automatic absent generation, attendance percentage calculation, and attendance analytics for both administrators and users.

1.2 Problem Statement

Traditional attendance management systems face several challenges including manual errors, time consumption, data redundancy, and attendance fraud. Although facial recognition systems automate attendance recording, they remain vulnerable to spoofing attacks where unauthorized individuals can mark attendance using photographs, mobile screens, replay videos, or similar fraudulent methods.

Furthermore, many attendance systems lack mechanisms for session-based attendance management, late attendance tracking, automatic absent generation, and attendance analytics. These limitations reduce the effectiveness and reliability of attendance monitoring.

Therefore, there is a need for a secure and intelligent attendance system capable of distinguishing genuine users from spoofing attempts while providing accurate attendance tracking, attendance percentage calculation, and automated attendance management.

1.3 Objectives of the Project

The primary objectives of the proposed system are:

1. To develop an automated attendance management system using facial recognition technology.
2. To detect and prevent spoofing attacks using a CNN-based liveness detection model.
3. To distinguish genuine faces from fake photographs, replay videos, and digital displays.
4. To improve attendance security and eliminate proxy attendance.
5. To automate attendance recording and storage using a MySQL database.
6. To provide real-time face detection and recognition.
7. To implement session-based attendance management using Morning and Evening attendance windows.

8. To identify and record Late attendance automatically.
9. To generate Absent records automatically for users who fail to mark attendance.
10. To calculate attendance percentage based on 120 working days.
11. To provide attendance analytics including Full Day, Half Day, Present, Late, and Absent statistics.
12. To generate attendance reports for administrators and users.

1.4 Scope of the Project

The scope of the proposed Anti-Spoofing Face Attendance System includes:

- Real-time face detection using a webcam.
- CNN-based liveness verification for spoof detection.
- Facial recognition of registered users.
- Automated attendance recording and management.
- Morning and Evening attendance sessions.
- Late attendance identification.
- Automatic absent generation for missed attendance sessions.
- Attendance percentage calculation based on 120 working days.
- User attendance dashboard and attendance history tracking.
- Administrator attendance monitoring and reporting.
- Secure storage of attendance records in a MySQL database.

The system is primarily designed for educational institutions and organizations where secure attendance management is required. Future enhancements can include cloud integration, mobile applications, multi-camera support, and advanced biometric authentication techniques.

1.5 Methodology and SDLC Model Adopted

The development of the proposed system follows the Software Development Life Cycle (SDLC) methodology.

Requirement Analysis

The requirements related to face recognition, anti-spoofing verification, attendance management, attendance analytics, and report generation were identified and analyzed.

System Design

The system architecture, database structure, user interface, attendance workflow, and CNN model architecture were designed.

Implementation

The system was implemented using Python and Django. OpenCV was used for face detection, Face Recognition Library for user identification, TensorFlow and Keras for CNN-based anti-spoofing, and MySQL for database management.

Testing

Individual modules including face detection, spoof detection, facial recognition, attendance management, and database operations were tested independently and collectively.

Deployment and Maintenance

The developed system can be deployed in educational institutions and organizations. Future maintenance includes updating user data, retraining models, and enhancing attendance analytics.

1.6 Organization of the Report

The project report is organized into ten chapters.

Chapter 1 introduces the project, objectives, scope, problem statement, and development methodology.

Chapter 2 presents the literature survey, existing systems, limitations, and comparative study.

Chapter 3 discusses system analysis including feasibility study, software requirements specification, and functional requirements.

Chapter 4 explains the system design including architecture, database design, UML diagrams, workflow, and user interface design.

Chapter 5 describes the implementation details, technology stack, modules, and code integration strategy.

Chapter 6 presents the testing methodology, test cases, bug tracking, and quality assurance activities.

Chapter 7 discusses the results obtained from the developed system and performance evaluation.

Chapter 8 provides conclusions, limitations, challenges faced, and future enhancements.

Chapter 9 contains references used during the development of the project.

Chapter 10 includes appendices such as project schedules, test reports, additional screenshots, source code repository details, and supporting documents.

CHAPTER 2

LITERATURE SURVEY

2.1 Review of Existing Systems

Face recognition technology has emerged as one of the most popular biometric authentication methods due to its contactless nature and ease of deployment. Researchers have proposed various techniques to improve face recognition accuracy and protect systems from spoofing attacks. Several studies have focused on liveness detection, deep learning approaches, and attendance automation.

Paper 1

Recent Advances in Biometric Security: A Case Study of Liveness Detection in Face Recognition (Ito et al., 2017)

The authors discussed the importance of liveness detection in biometric authentication systems. The study highlighted the vulnerability of face recognition systems to photo attacks and proposed deep learning techniques for detecting genuine users.

Methodology

- Deep Learning based Liveness Detection
- Feature Extraction
- Face Classification

Advantages

- Improved security
- Better spoof detection capability

Limitations

- High computational complexity
- Dataset dependent performance

Paper 2**A Review of Face Anti-Spoofing Methods for Face Recognition Systems (Anthony et al., 2021)**

This paper presented a detailed survey of anti-spoofing techniques including texture analysis, motion analysis, and deep learning-based methods.

Methodology

- Texture Analysis
- Motion Analysis
- CNN Based Detection

Advantages

- Comprehensive attack analysis
- Improved spoof detection

Limitations

- Reduced performance under real-world conditions
- High dependence on training data

Paper 3**Spoofing Detection in Face Recognition: A Review (Bagga & Singh, 2016)**

The authors reviewed different spoofing attacks and software-based detection mechanisms used in biometric systems.

Methodology

- Texture Feature Analysis
- Hardware and Software Approaches

Advantages

- Covers multiple attack scenarios
- Effective classification techniques

Limitations

- Limited deep learning implementation
- Lower accuracy against advanced attacks

Paper 4**Face Liveness Detection Using a Sequential CNN Technique (Mohamed et al., 2021)**

This study proposed a CNN-based liveness detection framework trained using the Celeb A-Spoof dataset.

Methodology

- Sequential CNN Architecture
- Deep Feature Extraction
- Binary Classification

Dataset

- CelebA-Spoof Dataset

Accuracy

- 87%

Advantages

- Automatic feature extraction
- Better generalization capability

Limitations

- Requires large training datasets
- Moderate accuracy

Paper 5**Face Anti-Spoofing Using CNN Classifier and Face Liveness Detection
(Hadiprakoso et al., 2020)**

The authors combined color texture analysis with CNN classification to distinguish real faces from spoof faces.

Methodology

- Color Space Transformation
- CNN Classification

Advantages

- Effective under different lighting conditions
- Robust against print attacks

Limitations

- Training intensive
- Sensitive to image quality

Paper 6**Anti-Spoofing Door Lock Using Face Recognition and Blink Detection (Ganjoo & Purohit, 2021)**

This work integrated blink detection with facial recognition to improve security.

Methodology

- Face Recognition
- Blink Detection

Accuracy

- 92.68%

Advantages

- Cost-effective implementation
- Simple architecture

Limitations

- Blink detection failures in poor lighting
- Vulnerable to advanced replay attacks

Paper 7**FaceTime: Deep Learning Based Face Recognition Attendance System (Arsenovic et al., 2017)**

The authors developed an attendance management system using deep learning and facial recognition techniques.

Methodology

- Face Detection
- Face Embedding Generation
- Attendance Recording

Accuracy

- 95.02%

Advantages

- Automated attendance process
- High recognition accuracy

Limitations

- No anti-spoofing mechanism
- Vulnerable to photo attacks

2.2 Limitations of Existing Approaches

Although significant progress has been made in facial recognition systems, several limitations still exist.

Vulnerability to Spoofing Attacks

Many systems can be deceived using:

- Printed photographs
- Mobile screen images
- Replay videos
- Digital displays

Lack of Liveness Detection

Traditional attendance systems focus only on recognition and fail to verify whether the detected face is genuine.

Environmental Sensitivity

Performance decreases due to:

- Poor lighting
- Pose variations
- Camera quality
- Background noise

High False Acceptance Rate

Unauthorized users may gain access through spoofing attacks.

Limited Attendance Analytics

Most systems do not provide:

- Late attendance detection
- Session-based attendance
- Attendance percentage calculation
- Full Day and Half Day attendance monitoring

2.3 Need for the Proposed System

The limitations of existing attendance systems create the need for a secure and intelligent attendance management solution.

The proposed Anti-Spoofing Face Attendance System addresses these issues by integrating CNN-based liveness detection with facial recognition and attendance analytics.

The proposed system provides:

- Real-time face detection
- CNN-based spoof detection
- Facial recognition
- Automated attendance recording
- Session-based attendance management
- Late attendance detection
- Automatic absent generation
- Attendance percentage calculation
- User and administrator dashboards

By verifying face authenticity before recognition, the system significantly reduces attendance fraud and improves overall reliability

2.4 Comparative Study

Table 2.1 Comparative Study of Existing Systems

Author	Technique	Accuracy	Limitation
Ito et al. (2017)	Deep Learning Liveness Detection	89%	High Computation
Bagga & Singh (2016)	Texture Analysis	85%	Sensitive to Lighting
Mohamed et al. (2021)	Sequential CNN	87%	Moderate Accuracy
Ganjoo & Purohit (2021)	Blink Detection + Recognition	92.68%	Replay Attacks
Arsenovic et al. (2017)	CNN Face Recognition	95.02%	No Anti-Spoofing
Proposed System	CNN Anti-Spoofing + Face Recognition + Attendance Analytics	97%	Requires Training Data

2.5 Summary

The literature survey shows that existing face recognition attendance systems provide automated attendance recording but remain vulnerable to spoofing attacks and often lack advanced attendance management features. To overcome these limitations, the proposed Anti-Spoofing Face Attendance System integrates CNN-based liveness detection with facial recognition. The system enhances security, prevents proxy attendance, supports session-based attendance management, and provides accurate attendance analytics and reporting.

CHAPTER 3

SYSTEM ANALYSIS

3.1 Feasibility Study

A feasibility study is conducted to determine whether the proposed system is practical, economical, and beneficial for implementation. It helps evaluate the viability of the project before development and deployment.

3.1.1 Technical Feasibility

Technical feasibility determines whether the required technology, software, and hardware resources are available for successful implementation.

The proposed Anti-Spoofing Face Attendance System is technically feasible because it utilizes widely available technologies such as Python, Django, OpenCV, TensorFlow, Keras, Face Recognition Library, NumPy, and MySQL. The system operates on a standard computer equipped with a webcam and does not require specialized hardware.

The CNN model used for anti-spoofing can be trained and deployed using commonly available machine learning frameworks. Face detection, face recognition, attendance analytics, and database management are implemented using open-source technologies, reducing technical complexity and development costs.

Technical Benefits

- Uses open-source software and libraries.
- Supports real-time face detection and recognition.
- Provides secure attendance management through anti-spoofing verification.
- Easy integration with educational and organizational environments.
- Scalable and maintainable architecture.

3.1.2 Economic Feasibility

Economic feasibility evaluates whether the benefits of the proposed system outweigh its development and operational costs.

The proposed system requires only:

- A computer system.
- A webcam.
- Open-source software libraries.
- MySQL database.

Since all major software components are open source, the overall implementation cost is very low.

Economic Benefits

- Low development and maintenance cost.
- Eliminates paper-based attendance records.
- Reduces administrative workload.
- Minimizes attendance fraud and proxy attendance.
- Improves attendance management efficiency.

Therefore, the proposed system is economically feasible.

3.1.3 Operational Feasibility

Operational feasibility determines whether the system can effectively function in the intended environment and satisfy user requirements.

The proposed system provides a simple web-based interface that allows administrators and users to interact with the system efficiently.

The administrator can:

- Register users.
- Delete users.
- View attendance reports.
- Monitor attendance percentages.

The user can:

- Log in securely.
- Mark attendance.
- View attendance history.
- Monitor attendance percentage.

The system is user-friendly and requires minimal technical knowledge for operation.

Operational Benefits

- Easy to use.
- Fast attendance recording.
- Reduced human intervention.
- Improved security and reliability.
- Efficient attendance monitoring.

Hence, the proposed system is operationally feasible.

3.1.4 Time and Cost Estimation

The project was developed following the Software Development Life Cycle (SDLC) methodology.

Table 3.1 Time Estimation

Activity	Duration
Requirement Analysis	1 Week
Literature Survey	1 Week
System Design	1 Week
Implementation	3 Weeks
Testing	1 Week
Documentation	1 Week

Table 3.2 Estimated Cost

Resource	Estimated Cost
Software Tools	Free (Open Source)
Development Platform	Existing System
Webcam	Existing Device
Database	Free
Documentation	Minimal
Total Cost	Low

The project can be implemented with minimal investment, making it highly cost-effective.

3.2 Software Requirements Specification (SRS)

Software Requirements Specification (SRS) describes the functional and non-functional requirements of the proposed system.

3.2.1 Purpose

The purpose of the proposed Anti-Spoofing Face Attendance System is to provide a secure and automated attendance management solution using facial recognition and CNN-based liveness detection. The system prevents proxy attendance, improves attendance accuracy, and automates attendance recording and monitoring.

3.2.2 Scope

The system performs the following operations:

- Face Detection
- Face Anti-Spoofing Verification
- Face Recognition
- User Registration
- Attendance Recording
- Attendance Percentage Calculation
- Automatic Absent Generation
- Late Attendance Detection
- Attendance Report Generation

The system is intended for schools, colleges, universities, offices, and organizations that require secure attendance management.

3.2.3 User Characteristics

Administrator

Responsibilities

- Register new users.
- Delete registered users.
- View attendance records.
- Monitor attendance percentages.
- Generate attendance reports.
- Manage attendance database.

Registered User

Responsibilities

- Log into the system.
- Mark attendance using face authentication.
- View attendance history.
- Check attendance percentage.
- Monitor Present, Late, Half Day, and Absent records.

3.2.4 Hardware Requirements

Table 3.3 Hardware Requirements

Component	Specification
Processor	Intel i3 or Above
RAM	4 GB Minimum
Hard Disk	500 GB
Webcam	Standard HD Camera
Display	Monitor
Internet	Optional

3.2.5 Software Requirements

Table 3.4 Software Requirements

Software / Component	Details
Operating System	Windows 10 or Above
Programming Language	Python 3.7.0
Frontend	HTML, CSS
Backend	Django
Database	MySQL
Deep Learning Framework	TensorFlow
CNN Development	Keras
Face Detection	OpenCV
Face Recognition	Face Recognition Library
Numerical Computation	NumPy
Data Processing	Pandas
IDE	Visual Studio Code
Browser	Google Chrome

Database Requirements

Database Name:

faceattendance

Users Table

Field Name	Data Type
person_id	VARCHAR
person_name	VARCHAR

Attendance Table

Field Name	Data Type
person_id	VARCHAR
attendance_date	VARCHAR
attendance_time	VARCHAR
status	VARCHAR
session_type	VARCHAR

The database stores user information and attendance records. Attendance records include Morning and Evening session details, attendance status, and attendance history for report generation.

3.3 Functional and Non-Functional Requirements

Functional Requirements

Functional requirements describe the operations and services provided by the system.

FR1 User Registration

The system shall allow the administrator to register new users by capturing facial data and storing face encodings for future recognition.

FR2 Face Detection

The system shall detect faces from live webcam video streams using OpenCV.

FR3 Anti-Spoofing Detection

The system shall classify detected faces as Real or Spoof using a Convolutional Neural Network (CNN) model.

FR4 Face Recognition

The system shall compare facial encodings with stored encodings and identify registered users.

FR5 User Authentication

The system shall allow only authenticated and recognized users to mark attendance.

FR6 Attendance Recording

The system shall automatically record attendance details including date, time, attendance status, and session type.

FR7 Session-Based Attendance Management

The system shall support Morning and Evening attendance sessions with predefined attendance windows.

FR8 Late Attendance Detection

The system shall automatically mark users as Late when attendance is recorded during the grace period.

FR9 Automatic Absent Generation

The system shall automatically generate absent records for users who fail to mark attendance within the allowed attendance sessions.

FR10 Attendance Percentage Calculation

The system shall calculate attendance percentage based on 120 working days.

FR11 Attendance Analytics

The system shall calculate and display:

- Full Day Attendance
- Half Day Attendance
- Present Count
- Late Count
- Absent Count
- Attendance Percentage

FR12 Attendance Report Generation

The system shall generate attendance reports for administrators and users.

FR13 User Management

The system shall allow administrators to delete user records and associated face encodings when required.

Non-Functional Requirements

Non-functional requirements describe the quality attributes and performance expectations of the system.

Performance Requirements

- Face detection should occur in real time.
- Attendance marking should complete within a few seconds.
- The system should support continuous webcam operation without significant delay.

Security Requirements

- Attendance shall be recorded only after successful anti-spoofing verification.
- Unauthorized users shall not be allowed to mark attendance.
- Attendance records shall be protected from unauthorized modification.
- User facial data shall be stored securely.

Reliability Requirements

- The system should operate continuously without crashes.
- Face recognition results should remain consistent under normal operating conditions.
- Attendance records should be stored accurately in the database.

Usability Requirements

- The user interface should be simple and easy to understand.
- Users should be able to mark attendance with minimal interaction.
- Administrators should be able to manage attendance records efficiently.

Maintainability Requirements

- System modules should be easy to update and maintain.
- The CNN model should support retraining with new datasets.
- Database structures should allow future enhancements.

Scalability Requirements

- Additional users can be registered without affecting system performance.
- Larger datasets can be used for future model training.
- Additional attendance features can be integrated in future versions.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The proposed Anti-Spoofing Face Attendance System consists of multiple integrated modules that work together to provide secure and automated attendance management. The system captures live facial images through a webcam, verifies face authenticity using a CNN-based anti-spoofing model, performs facial recognition, and records attendance automatically in the database.

The major modules of the system are:

User Registration Module

This module allows the administrator to register new users by entering user details and capturing facial information. The extracted facial encodings are stored for future recognition.

Face Detection Module

The Face Detection Module uses OpenCV to detect human faces from live webcam video streams before further processing.

Anti-Spoofing Module

The Anti-Spoofing Module utilizes a Convolutional Neural Network (CNN) model to determine whether the detected face is genuine or spoofed. This prevents unauthorized attendance marking using photographs, videos, or digital displays.

Face Recognition Module

The Face Recognition Module compares live facial encodings with stored facial encodings and identifies registered users.

Attendance Management Module

This module records attendance information including date, time, session type, and attendance status. It also manages late attendance detection and automatic absent generation.

Attendance Analytics Module

The Attendance Analytics Module calculates:

- Full Day Attendance
- Half Day Attendance
- Present Count
- Late Count
- Absent Count
- Attendance Percentage

The percentage is calculated based on 120 working days.

Database Module

The MySQL database stores user information and attendance records. It provides secure data storage and retrieval for attendance reporting and analysis.

The architecture ensures accurate attendance recording, secure authentication, and reliable attendance monitoring.

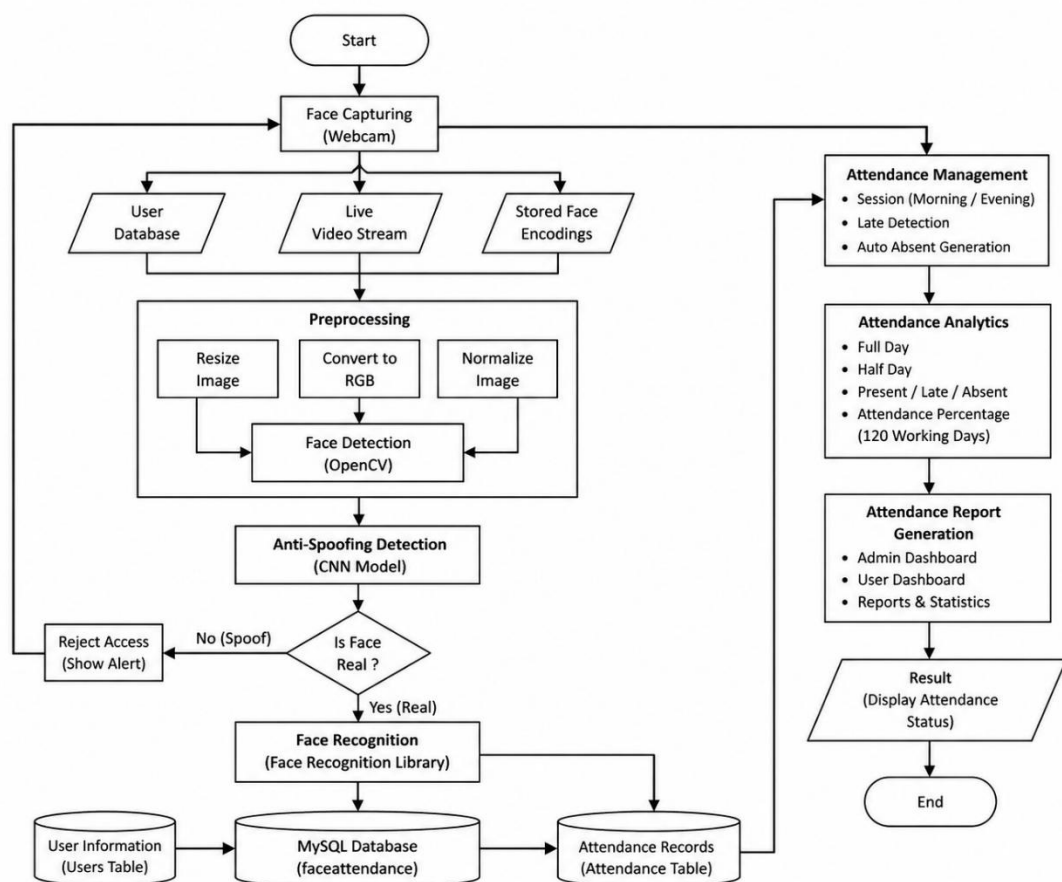


Fig. 4.1: System Architecture

4.2 Database Design and Workflow

The proposed system uses a MySQL database named faceattendance to store user information and attendance records. The database is designed to maintain data integrity, reduce redundancy, and support efficient attendance management.

The Users table stores registered user details, while the Attendance table stores attendance records including attendance date, attendance time, session type, and attendance status. A one-to-many relationship exists between the Users table and the Attendance table, as a single user can have multiple attendance records.

The workflow of the proposed system begins with live face capture through a webcam. The captured image is processed using OpenCV for face detection. The detected face is then verified using a CNN-based anti-spoofing model. If the face is identified as genuine, facial recognition is performed using stored face encodings. Once the user is recognized, attendance details are recorded in the database. The attendance management module then updates attendance statistics and generates reports for users and administrators.

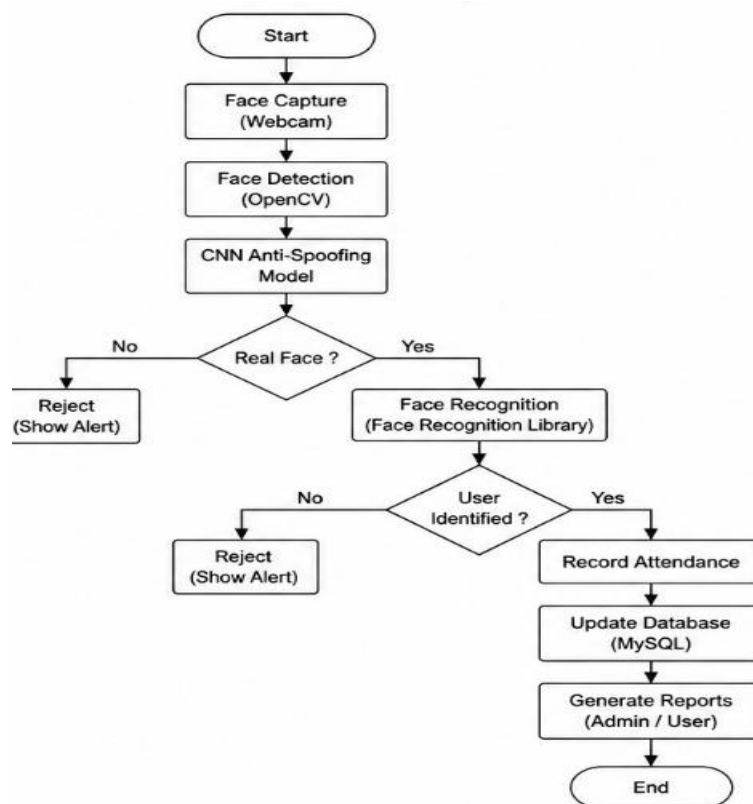


Fig. 4.2 Workflow Diagram

4.3 UML Diagrams (Use Case, Class, Sequence, Activity)

Unified Modeling Language (UML) diagrams are used to represent the structure, behavior, and interaction of different components within the Anti-Spoofing Face Attendance System. UML diagrams provide a visual representation of system functionality and help in understanding the overall system design.

4.3.1 Use Case Diagram

The Use Case Diagram illustrates the interaction between the Administrator, User, and the proposed attendance system. The Administrator performs tasks such as dataset management, CNN model training, attendance monitoring, and report generation. The User interacts with the system through face authentication and attendance marking. The diagram provides a high-level representation of the system functionalities and user interactions.

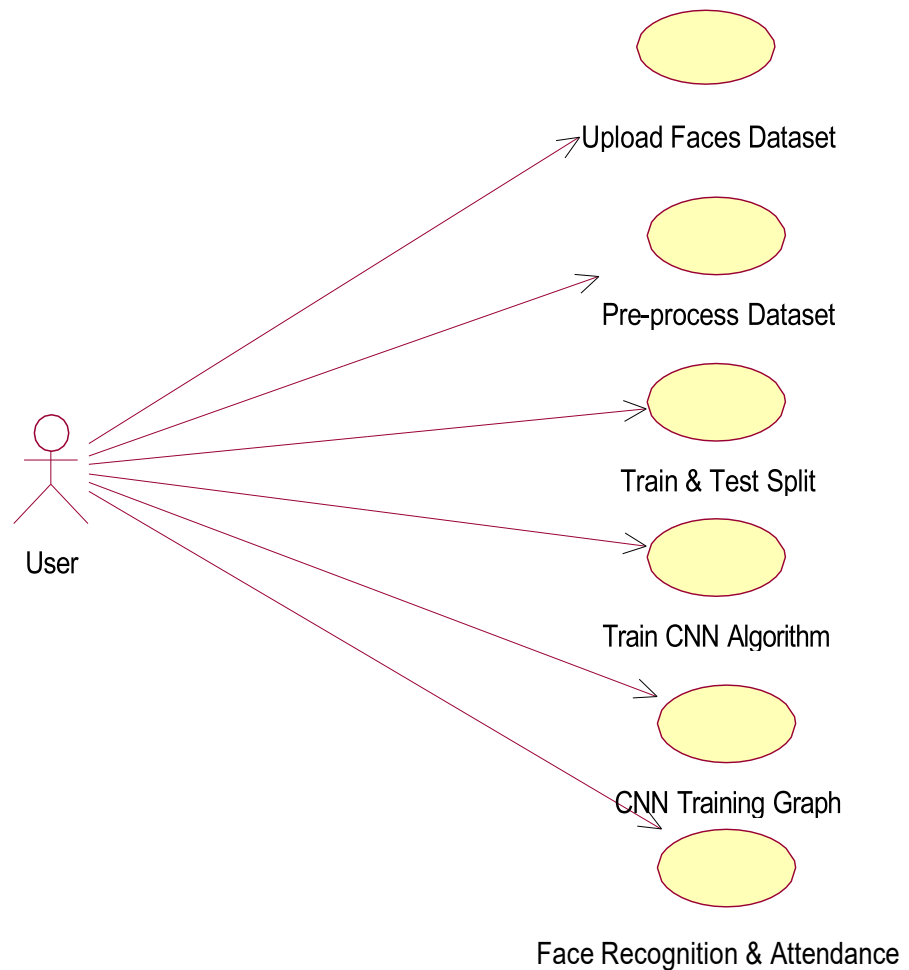


Figure 4.5: Use Case Diagram

4.3.2 Class Diagram

The Class Diagram represents the static structure of the system. It consists of classes such as FaceAttendance, FaceDetector, and FaceSpoofing. These classes contain attributes and methods that perform face detection, spoof detection, face recognition, attendance recording, and database operations. The relationships among classes define how different modules communicate within the system.

The FaceAttendance class contains attributes including filename, X, Y, X_train, X_test, y_train, y_test, cnn_model, labels, sc, names, embedding, and attendance. Its methods include Upload Faces Dataset, Pre-process Dataset, Train and Test Split, Train CNN Algorithm, CNN Training Graph, and Face Recognition and Attendance. The FaceDetector class includes modelFile, configFile, and model attributes, with methods get_face_detector, find_faces, and draw_faces. The FaceSpoofing class manages face_model, clf, cap, sample_number, count, and measures.

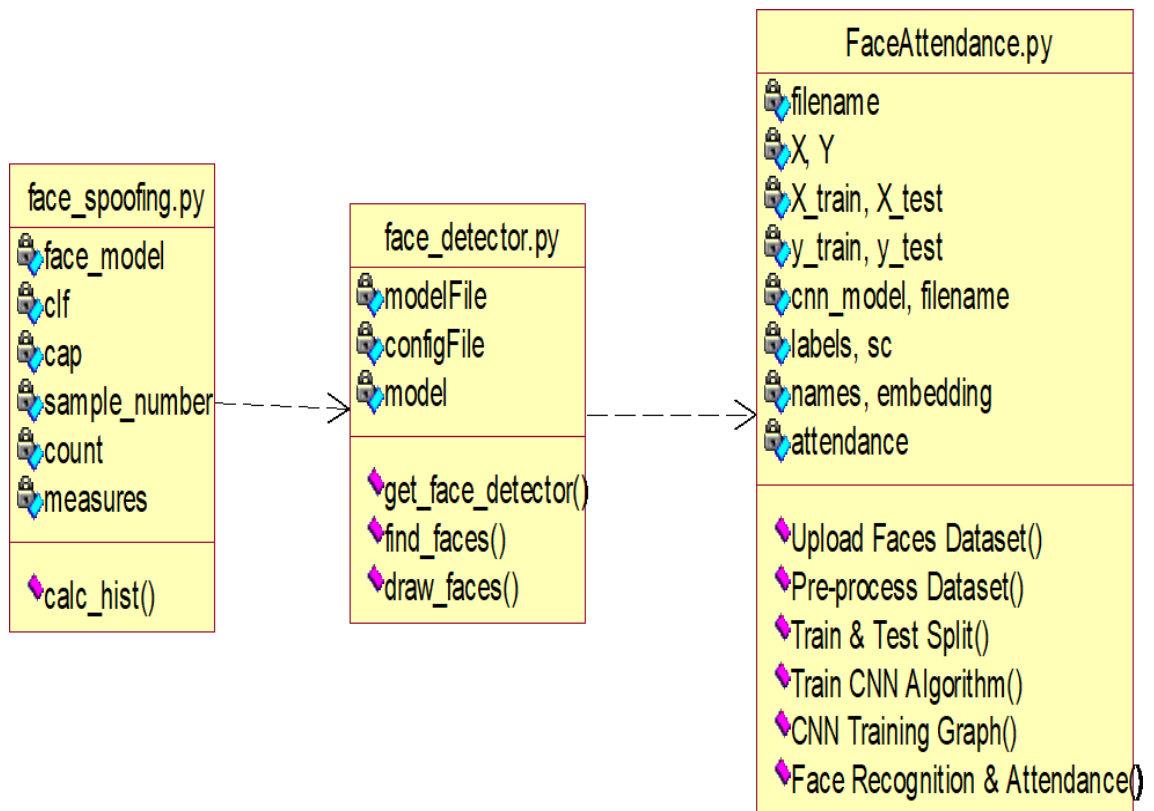


Figure 4.6: Class Diagram

4.3.3 Sequence Diagram

The Sequence Diagram describes the order of interactions between system components during attendance marking. The webcam captures the user's face and sends the image to the face detection module. The detected face is verified by the CNN anti-spoofing model. If the face is classified as genuine, the recognition module identifies the user and updates the attendance database. The sequence diagram clearly shows the message flow between different modules.

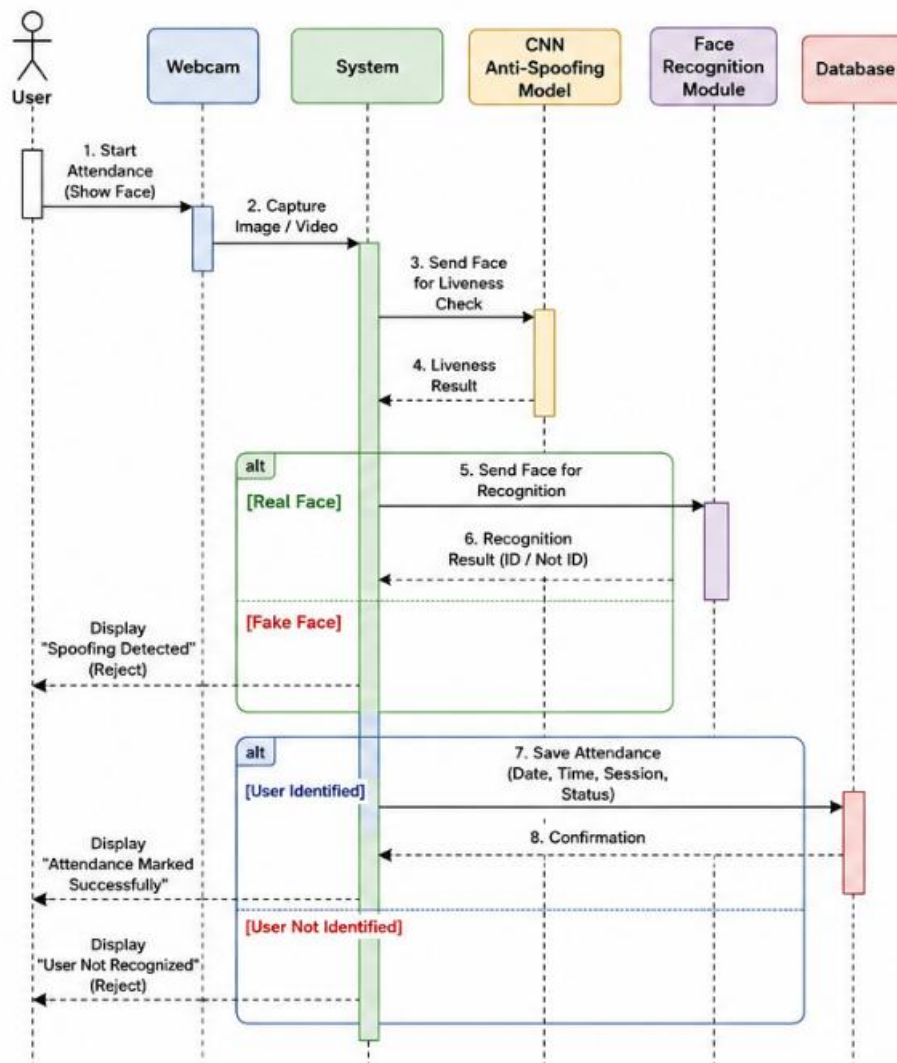


Figure 4.7: Sequence Diagram

4.3.4 Activity Diagram

The Activity Diagram illustrates the workflow of the Anti-Spoofing Face Attendance System. The process begins with image capture through a webcam. The system then detects the face and performs anti-spoofing verification. If the face is genuine, face recognition is executed and attendance is recorded in the database. If a spoof face is detected, the system rejects the attendance request. The activity diagram helps visualize the complete flow of operations from image acquisition to attendance storage.

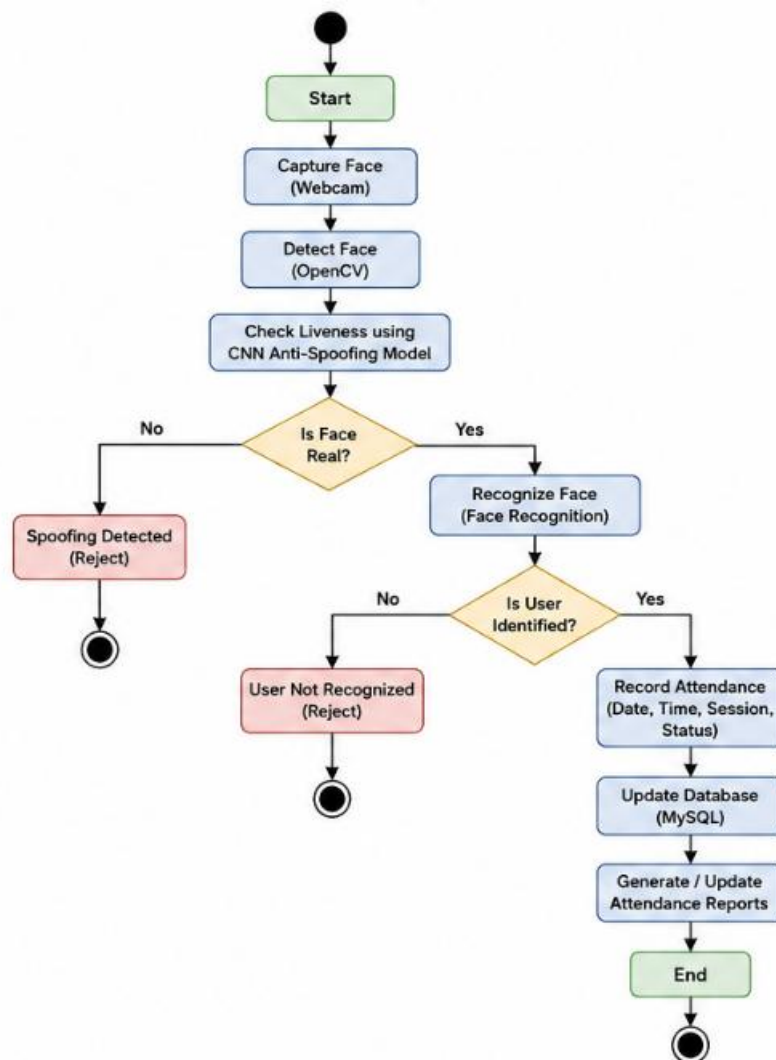


Figure 4.8: Activity Diagram

4.4 User Interface Design

The User Interface (UI) of the Anti-Spoofing Face Attendance System is designed to provide a simple, interactive, and user-friendly environment for administrators and users. The interface enables efficient dataset management, CNN model training, face recognition, attendance recording, and report generation. The GUI is developed using HTML, CSS, and Django, allowing users to perform all operations through graphical controls without requiring technical expertise.

4.4.1 Home Page

The Home Page serves as the entry point of the Anti-Spoofing Face Attendance System. It displays the system title and provides a Login button for accessing the system. The interface is designed with a clean and modern layout featuring a technology-themed background that reflects the AI-based nature of the system.

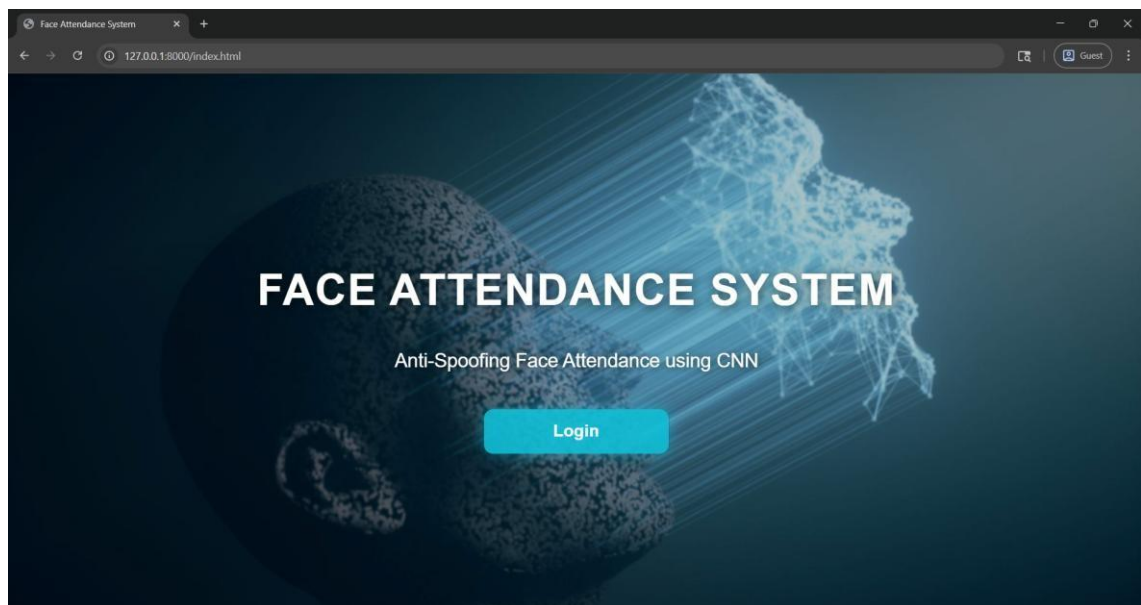


Figure 4.9: Home Page

4.4.2 Login Interface

The Login Interface allows administrators and registered users to securely access the system. The page provides two options: Admin Login and User Login. Administrators can access the admin panel to manage users and attendance records, while users can access the attendance marking module to record their attendance.

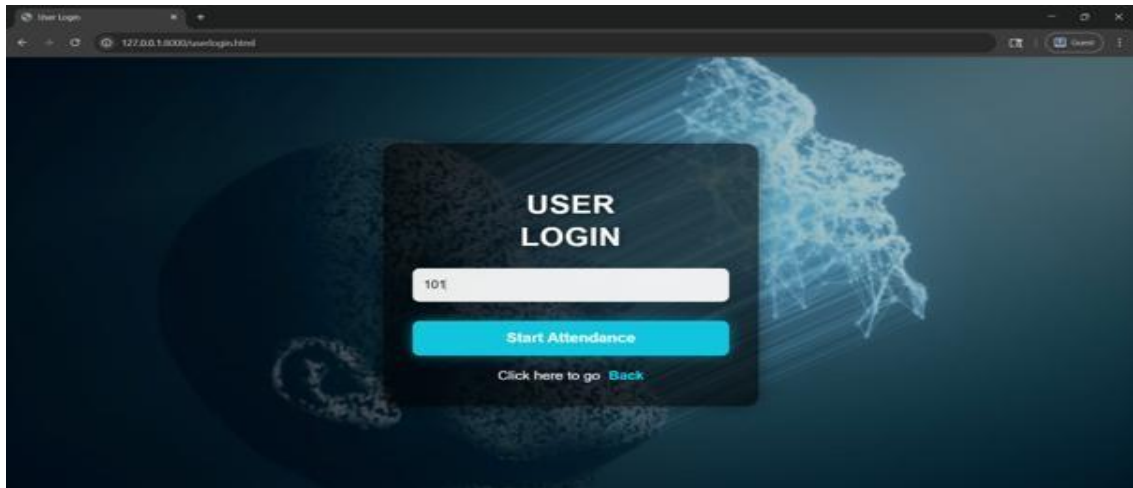


Figure 4.10: Login Interface

4.4.3 Administrator Dashboard

The Administrator Dashboard provides modules for user registration, attendance viewing, and user management. The admin login page requires username and password credentials for secure access. This interface enables administrators to efficiently control system operations and monitor attendance records across the organization.



Figure 4.11: Administrator Dashboard

4.4.4 User Registration Interface

The User Registration Interface allows administrators to register new users by entering the Person ID and Person Name. The system captures facial data from the webcam during registration and stores the facial encodings for future recognition and attendance verification. Users are advised to ensure their face is clearly visible during registration.

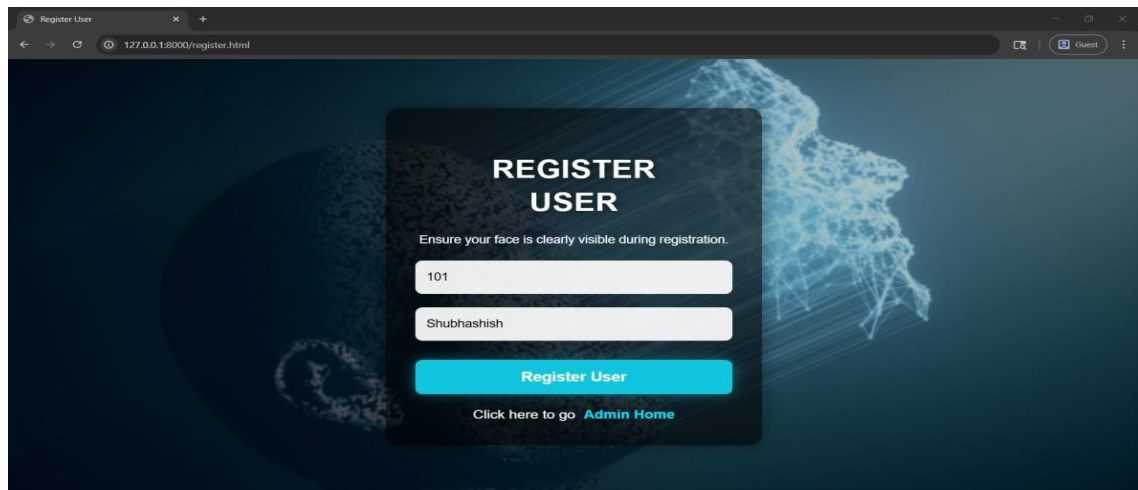


Figure 4.12: User Registration Interface

4.4.5 Attendance Interface

The Attendance Interface enables users to initiate the attendance marking process. The user enters their Person ID and clicks the Start Attendance button to activate the webcam. The system then begins face detection, anti-spoofing verification, and user recognition automatically. Upon successful authentication, attendance is recorded in the database.

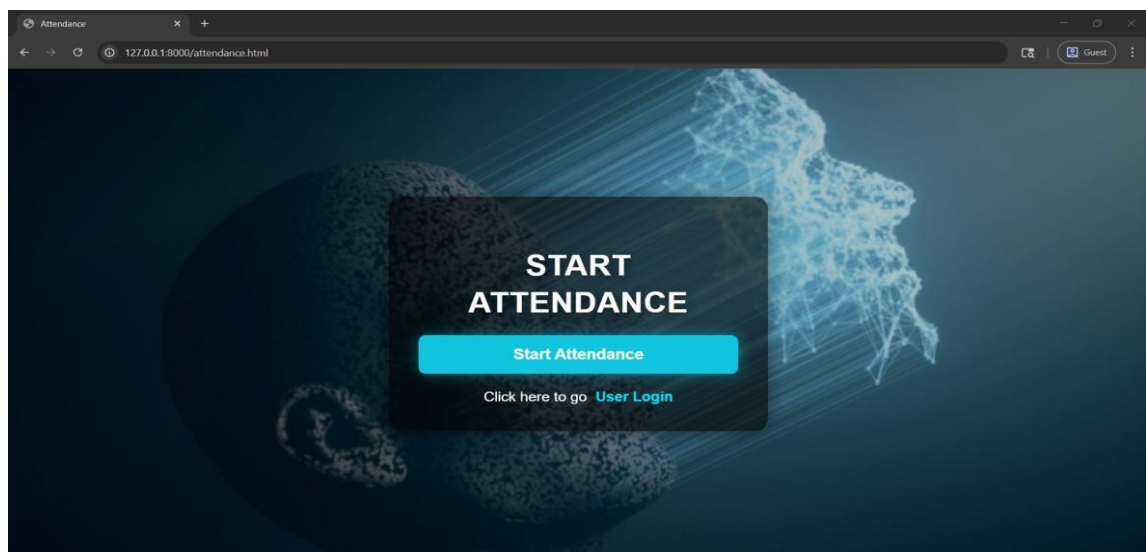


Figure 4.13: Attendance Interface

4.5 Design Standards Followed

The design and development of the proposed Anti-Spoofing Face Attendance System follow standard software engineering principles and internationally accepted guidelines to ensure reliability, maintainability, security, and usability.

IEEE Standards

The system development follows the principles of the IEEE Software Engineering Standards, particularly in requirements analysis, system design, implementation, testing, and documentation. UML diagrams and Software Requirement Specifications (SRS) are prepared according to standard software engineering practices.

ISO Quality Standards

The system design considers the quality characteristics defined in ISO 25010 Software Quality Model, including Functional Suitability, Reliability, Security, Maintainability, Performance Efficiency, and Usability. These quality factors ensure that the system performs accurately and efficiently under different operating conditions.

UML Design Standards

Unified Modeling Language (UML) standards are followed to represent the structure and behavior of the system. Use Case, Class, Sequence, and Activity Diagrams are used to model system functionality and interactions.

Database Design Standards

The MySQL database is designed using normalization principles to minimize data redundancy and maintain data integrity. Primary and foreign key relationships are utilized to establish proper connections between system entities.

User Interface Design Principles

The graphical user interface is developed with simplicity and usability in mind. The interface provides easy navigation, consistent screen layouts, clear labels, and user-friendly controls to improve the overall user experience.

Security Standards

The system incorporates CNN-based anti-spoofing techniques and user authentication mechanisms to improve attendance security and prevent unauthorized access. Attendance records are stored securely in the database to maintain data confidentiality and integrity.

4.6 Safety and Risk Mitigation Measures

The Anti-Spoofing Face Attendance System is designed with appropriate safety and security measures to ensure accurate attendance recording and protection against fraudulent activities. Various risk mitigation techniques are incorporated to improve system reliability and maintain data integrity.

Prevention of Spoofing Attacks

The system utilizes a Convolutional Neural Network (CNN) based anti-spoofing mechanism to distinguish genuine faces from fake images, mobile screen displays, and replay attacks. This prevents unauthorized users from marking attendance using spoofing techniques.

User Authentication

Only registered users whose facial data is available in the system database are allowed to mark attendance. Unrecognized users are automatically rejected, thereby reducing the possibility of unauthorized access.

Attendance Data Security

Attendance records are stored securely in the MySQL database. Proper database management practices are followed to prevent data loss, duplication, and unauthorized modifications.

Duplicate Attendance Prevention

The system verifies attendance records before insertion into the database to ensure that multiple attendance entries are not recorded for the same user on the same day and session.

Error Handling and Recovery

The application includes error handling mechanisms to manage situations such as webcam failure, missing face detection, invalid user input, and database connectivity issues. Appropriate messages are displayed to guide users and administrators.

Data Integrity

All user information and attendance records are validated before storage. This helps maintain consistency and accuracy of data throughout the system.

System Reliability

The modular architecture of the system ensures stable operation and simplifies maintenance. Individual modules such as face detection, anti-spoofing verification, face recognition, and attendance management operate independently, reducing the impact of failures in any single component.

Privacy Protection

The system stores only the information required for attendance management. Facial data and attendance records are maintained securely and are accessible only to authorized administrator

CHAPTER 5

IMPLEMENTATION

5.1 Technology Stack

The Anti-Spoofing Face Attendance System is developed using modern software technologies and machine learning frameworks. These technologies support face detection, anti-spoofing verification, face recognition, database management, and user interface development.

Table 5.1: Technology Stack Used

Technology	Purpose
Python	Core Programming Language
OpenCV	Face Detection and Image Processing
TensorFlow	Deep Learning Framework
Keras	CNN Model Development
NumPy	Numerical Computations
Pandas	Data Processing
Face Recognition Library	Facial Encoding and Recognition
Django	Backend Web Framework
MySQL	Attendance Database Management
HTML, CSS	Frontend User Interface
Matplotlib	Performance Graph Visualization
Visual Studio Code	Integrated Development Environment

5.2 Module-wise Implementation

The proposed system is divided into several modules to improve maintainability and functionality. Each module performs a specific role in the overall attendance management workflow.

Dataset Management Module

This module is responsible for collecting, storing, and managing user facial images. The administrator uploads a dataset of face images organized by user identity. The uploaded dataset is then prepared for preprocessing and CNN model training.

Preprocessing Module

The preprocessing module converts facial images into YCrCb and CIE Luv* colour spaces. Histogram features are extracted from each image to improve spoof detection accuracy. The preprocessing step normalizes pixel values and resizes images to a standard format suitable for CNN input.

CNN Anti-Spoofing Module

This module trains a Convolutional Neural Network using the extracted histogram features from real and spoof face images. The CNN model consists of convolutional layers, pooling layers, dropout layers, and dense layers. The model learns to distinguish genuine faces from spoofed faces by classifying them into Real Face and Spoof Face categories. The trained model is saved for use during real-time attendance marking.

Face Detection Module

OpenCV-based face detection techniques are used to locate facial regions from webcam frames. The system uses a Caffe-based face detector model that identifies the bounding box coordinates of detected faces. Only detected faces are passed to the anti-spoofing and recognition modules.

Face Recognition Module

The face recognition module uses the face_recognition library to extract 128-dimensional facial encodings from detected faces. These encodings are compared with stored encodings of registered users using Euclidean distance measurement. The user whose encoding is closest to the detected face encoding is identified as the recognized user.

Attendance Management Module

Once a user is successfully authenticated, attendance details including Person ID, date, time, session type, and attendance status are automatically stored in the MySQL database. The module handles Morning and Evening session attendance windows. Late attendance is automatically detected based on predefined time thresholds. The module also generates absent records for users who miss attendance sessions.

Administration Module

The administrator can register new users by capturing facial images, delete registered users from the system, and view attendance records through the administrative interface. The admin panel provides complete control over user management and attendance monitoring.

5.3 Code Integration Strategy

The implementation follows a modular integration strategy where each component operates independently and communicates through well-defined interfaces. The integration process ensures that data flows correctly between modules without causing errors or data loss.

The integration process follows these steps:

1. User data is collected and stored in the database during registration.
2. Face images are pre-processed for feature extraction using colour space transformation and histogram computation.
3. The CNN model is trained using the prepared dataset of real and spoof face images.
4. Live webcam captures facial images during attendance marking.
5. The face detection module extracts facial regions from the captured frame.
6. The anti-spoofing module verifies face authenticity using the trained CNN model.
7. The face recognition module identifies the user by comparing facial encodings.
8. The attendance module stores attendance details in the MySQL database.
9. Attendance reports are generated and displayed for administrators and users.

This modular approach simplifies maintenance, testing, and future enhancements.

5.4 Sample Code Snippets

The following code snippets illustrate the key implementation components of the Anti-Spoofing Face Attendance System.

Face Detection Using OpenCV

```
faces = detector.detectMultiScale(gray, scaleFactor=1.3, minNeighbors=5)
```

The above code detects facial regions from captured webcam images using a cascade classifier.

CNN Model Prediction

```
prediction = model.predict(features)
```

The trained CNN model predicts whether the detected face is genuine or spoofed by returning a classification result.

Attendance Storage in MySQL

```
cursor.execute("INSERT INTO attendance (person_id, date, time, status) VALUES (%s,%s,%s,%s)", (pid, date, time, status))
```

The attendance information is stored in the MySQL database after successful authentication.

Face Recognition Using Encodings

```
matches = face_recognition.compare_faces(known_encodings, face_encoding)
```

The system compares extracted facial encodings with registered user encodings to identify users.

Database Connection

```
conn = mysql.connector.connect(host="localhost", user="root", password="", database="faceattendance")
```

The above code establishes a connection between the application and the MySQL database.

5.5 Coding Standards Followed

The development of the Anti-Spoofing Face Attendance System follows standard coding practices to improve readability, maintainability, and reliability.

Naming Conventions

Meaningful variable names, function names, and class names are used throughout the application to improve code clarity and readability. All identifiers follow the Python PEP 8 naming convention

Modular Programming

The application is divided into independent modules such as preprocessing, spoof detection, recognition, attendance management, and database management. Each module handles a specific functionality and can be maintained or updated independently.

Code Reusability

Reusable functions are implemented to avoid code duplication and simplify maintenance. Common operations such as database connection, face detection, and image processing are encapsulated in reusable functions.

Error Handling

Exception handling mechanisms are incorporated to manage runtime errors, invalid inputs, and database connectivity issues. Try-except blocks are used to catch and handle errors gracefully without crashing the application.

Documentation

Comments and docstrings are included wherever necessary to explain complex logic, function parameters, and module purposes. This improves code understanding and facilitates future maintenance.

Database Security

Parameterized SQL queries are used to reduce the risk of SQL injection and ensure secure database operations. User inputs are validated before processing to prevent malicious data from entering the system.

CHAPTER 6

TESTING

6.1 Testing Strategy

Testing is a systematic process used to verify whether the developed Anti-Spoofing Face Attendance System satisfies the specified requirements and performs as expected. The testing process was conducted to ensure accuracy in face detection, anti-spoofing verification, face recognition, database operations, and attendance management.

The objectives of testing are:

- To verify correctness of each individual module.
- To identify and eliminate software defects.
- To validate attendance recording functionality.
- To evaluate system reliability and performance.
- To ensure secure operation against spoofing attacks.

The testing process included Unit Testing, Integration Testing, and System Testing. All modules were tested individually and then collectively to verify system-level functionality.

6.2 Unit Testing

Unit Testing is performed on individual modules independently to verify their functionality before integration. Each module is tested with valid and invalid inputs to ensure correct behavior.

Table 6.1: Unit Testing Results

Module	Purpose	Result
Admin Login	Verify administrator authentication	Pass
User Registration	Verify user registration process	Pass
Face Detection	Verify face detection functionality	Pass
CNN Anti-Spoofing	Verify spoof detection accuracy	Pass
Face Recognition	Verify user identification	Pass
Attendance Module	Verify attendance recording	Pass
View Attendance	Verify attendance retrieval	Pass
Delete User	Verify user deletion process	Pass
Attendance Analytics	Verify attendance percentage calculation	Pass

All modules performed successfully during unit testing. No critical defects were identified during this phase.

6.3 Integration Testing

Integration Testing verifies the interaction between multiple modules after unit testing is completed. It ensures that data flows correctly between modules and that combined modules produce the expected results.

The following module integrations were tested:

- User Registration to Database Storage
- Face Detection to CNN Anti-Spoofing
- CNN Anti-Spoofing to Face Recognition
- Face Recognition to Attendance Recording
- Attendance Module to MySQL Database

Table 6.2: Integration Testing Results

Integrated Modules	Expected Result	Result
Registration and Database	User data stored correctly	Pass
Detection and Anti-Spoofing	Real or Spoof classification	Pass
Anti-Spoofing and Recognition	Only real faces are recognized	Pass
Recognition and Attendance	Attendance marked correctly	Pass
Attendance and Database	Data stored successfully	Pass
Analytics and Database	Correct percentage calculated	Pass

The integrated modules exchanged information correctly without errors. All integration test cases produced expected results.

6.4 System Testing

System Testing evaluates the complete application as a whole under real operating conditions. The full system is tested to ensure all functionalities work correctly together.

The following functionalities were tested:

- Administrator Login
- User Registration
- User Login
- Face Detection from Webcam
- Anti-Spoofing Detection with Mobile Display Image
- Face Recognition of Registered Users
- Attendance Marking for Authenticated Users
- Attendance Viewing and Report Generation
- User Deletion

Table 6.3: System Testing Results

Test Scenario	Expected Output	Result
Valid Admin Login	Dashboard Opened Successfully	Pass
User Registration	User Added Successfully	Pass
Genuine Face Detection	Attendance Marked	Pass
Spoof Face Detection	Attendance Rejected	Pass
Unknown User Detection	Unknown User Message	Pass
View Attendance Records	Attendance Displayed	Pass
Delete User	User Removed Successfully	Pass
Attendance Percentage	Correct Calculation Displayed	Pass

6.5 Test Cases and Results

The following test cases document the detailed testing conducted for each major functionality of the system.

Table 6.4: TC01 - Dataset Upload

Field	Details
Test Case ID	TC01
Module	Dataset Upload
Objective	Verify dataset loading functionality
Input	Face image dataset organized by user
Expected Output	Dataset loaded and stored successfully
Actual Output	Dataset loaded and stored successfully
Result	Pass

Table 6.5: TC02 - Image Preprocessing

Field	Details
Test Case ID	TC02
Module	Image Preprocessing
Objective	Verify preprocessing operation and feature extraction
Input	Uploaded dataset of face images
Expected Output	Images pre-processed and features extracted successfully
Actual Output	Images pre-processed and features extracted successfully
Result	Pass

Table 6.6: TC03 - CNN Training

Field	Details
Test Case ID	TC03
Module	CNN Anti-Spoofing Training
Objective	Verify model training accuracy and convergence
Input	Pre-processed dataset of real and spoof images
Expected Output	CNN model trained and saved successfully
Actual Output	CNN model trained with 97% accuracy
Result	Pass

Table 6.7: TC04 - Face Detection

Field	Details
Test Case ID	TC04
Module	Face Detection
Objective	Detect face from live webcam stream
Input	Live webcam image with user in frame
Expected Output	Face detected with bounding box
Actual Output	Face detected successfully
Result	Pass

Table 6.8: TC05 - Anti-Spoofing Detection

Field	Details
Test Case ID	TC05
Module	Anti-Spoofing Detection
Objective	Detect spoof image presented via mobile screen
Input	Mobile display image of a face
Expected Output	Spoof detected, attendance rejected
Actual Output	Spoof detected, attendance rejected
Result	Pass

Table 6.9: TC06 - Face Recognition

Field	Details
Test Case ID	TC06
Module	Face Recognition
Objective	Recognize registered user from live face
Input	Live registered face presented to webcam
Expected Output	User identified correctly
Actual Output	User identified correctly
Result	Pass

Table 6.10: TC07 - Attendance Marking

Field	Details
Test Case ID	TC07
Module	Attendance Marking
Objective	Record attendance for authenticated user
Input	Authenticated and recognized user
Expected Output	Attendance stored in database
Actual Output	Attendance stored successfully
Result	Pass

Table 6.11: TC08 - Unknown User Detection

Field	Details
Test Case ID	TC08
Module	Unknown User Detection
Objective	Verify system handles unregistered users
Input	Unregistered face presented to webcam
Expected Output	Unknown User message displayed
Actual Output	Unknown User message displayed
Result	Pass

6.6 Bug Reporting and Tracking

Bug reporting and tracking is an important activity performed during software testing to identify, document, and resolve defects in the system. During the development of the Anti-Spoofing Face Attendance System, various issues related to face detection, database connectivity, attendance recording, and spoof detection were identified and corrected.

Table 6.12: Bug Tracking Report

Bug ID	Description	Status
B001	Face detection failure under poor lighting conditions	Fixed
B002	Duplicate attendance entries generated for the same user	Fixed
B003	Database connection error during attendance storage	Fixed
B004	Incorrect face recognition for similar facial features	Fixed
B005	Spoof detection threshold required optimization	Fixed
B006	Attendance percentage calculation error for partial sessions	Fixed

The identified bugs were analyzed and resolved during testing. The corrective actions improved system accuracy, reliability, and overall performance.

6.7 Quality Assurance Standards

Quality Assurance (QA) ensures that the developed software satisfies functional requirements, performance expectations, and security standards. The Anti-Spoofing Face Attendance System was developed following standard software engineering practices to ensure software quality.

Reliability

The system consistently performs face detection, anti-spoofing verification, face recognition, and attendance recording without major failures.

Accuracy

CNN-based anti-spoofing and facial recognition modules provide accurate authentication results and reduce false attendance marking.

Security

The system prevents spoofing attacks and allows attendance recording only for authenticated users. Attendance data is securely stored in the MySQL database.

Usability

The graphical user interface provides a simple and user-friendly environment for administrators and users.

Maintainability

The modular architecture enables easy maintenance, debugging, and future enhancements without affecting the overall system.

Performance

The system performs real-time face detection and attendance recording with minimal response delay, ensuring smooth operation during attendance sessions.

CHAPTER 7

RESULT and DISCUSSION

7.1 Output Screenshots

The following screenshots demonstrate the key outputs and functionalities of the developed Anti-Spoofing Face Attendance System.

7.1.1 Home Page

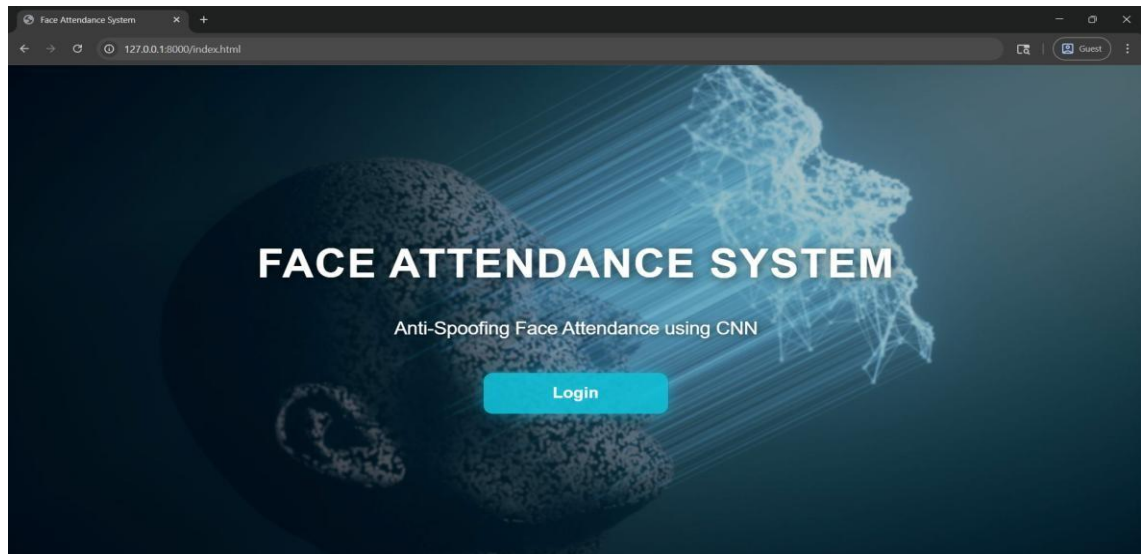


Figure 7.1: Home Page

The home page provides a clean interface with the system title and a Login button for accessing the system functionalities.

7.1.2 Login Interface

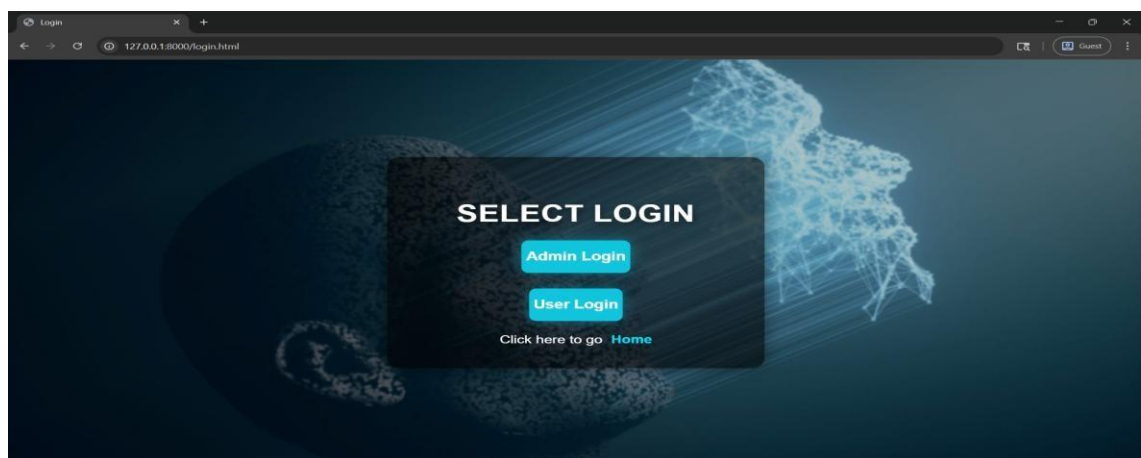


Figure 7.2: Login Interface

The login page allows administrators and users to access the system securely by selecting the appropriate login type.

7.1.3 Administrator Dashboard



Figure 7.3: Administrator Dashboard

The administrator dashboard provides options for user registration, attendance viewing, and user management. Administrators can control all system operations from this interface.

7.1.4 User Registration

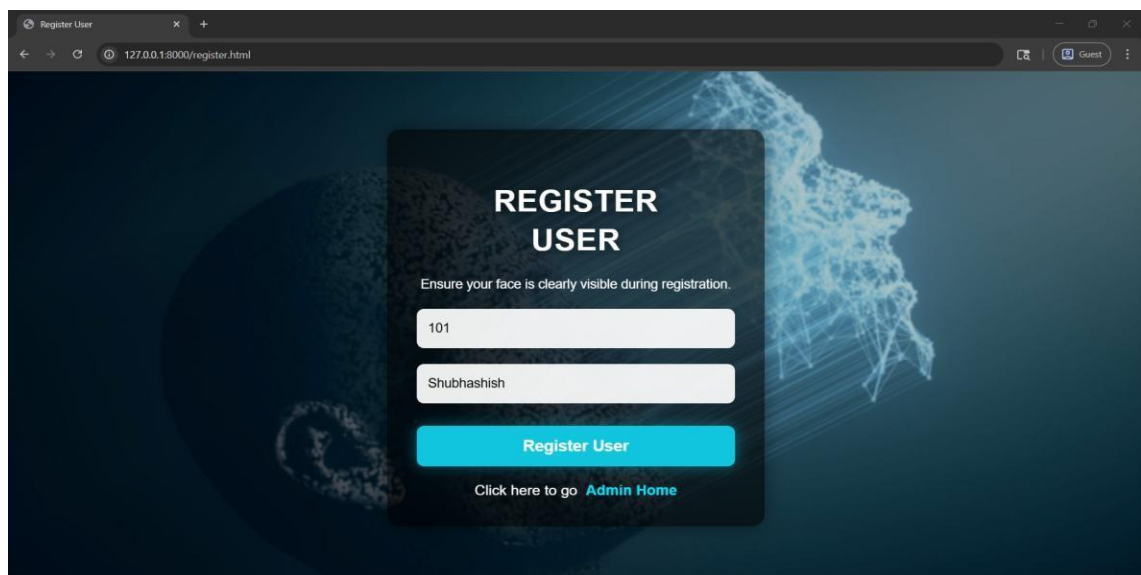


Figure 7.4: User Registration Interface

The system successfully registers a new user by capturing facial images and storing face encodings. The administrator enters the Person ID and Name to complete registration.

7.1.5 Attendance Interface

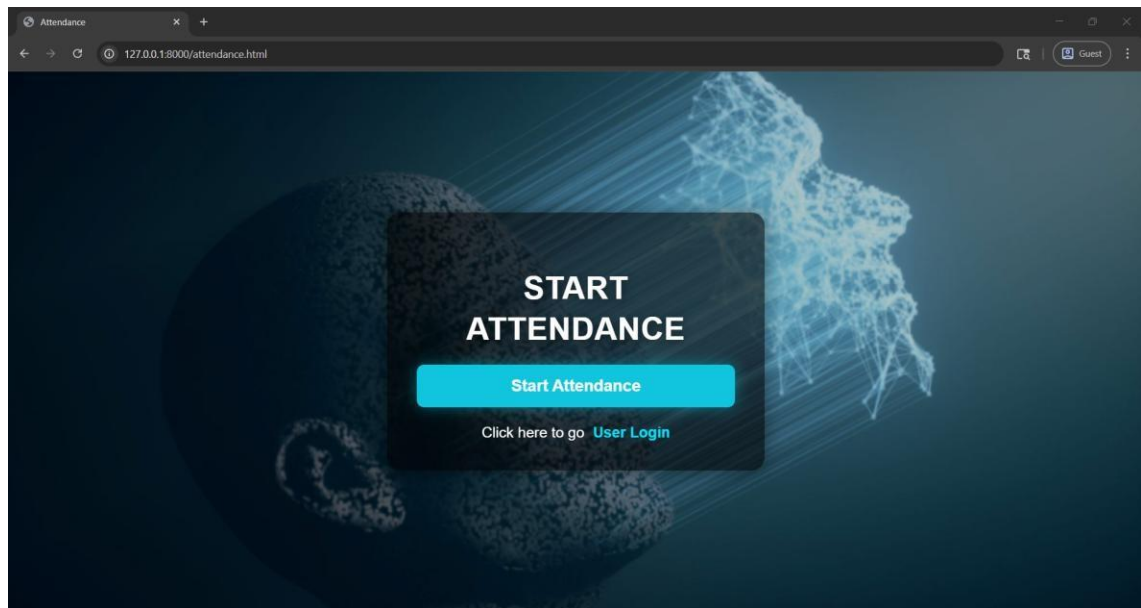


Figure 7.5: Attendance Interface

The attendance interface activates the webcam and initiates the face authentication process when the user clicks the Start Attendance button.

7.1.6 Successful Face Recognition

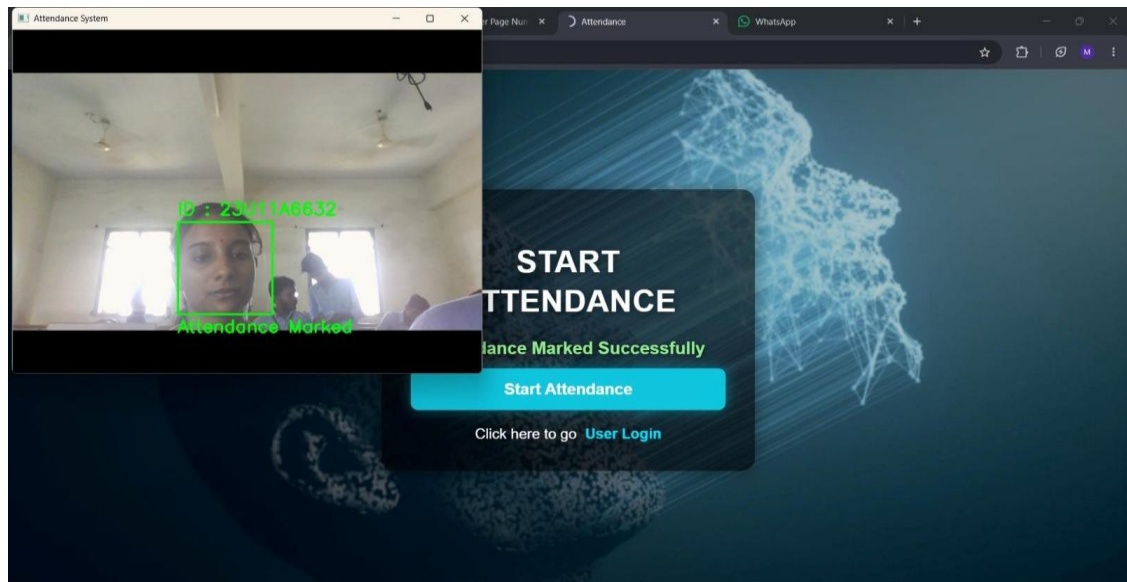
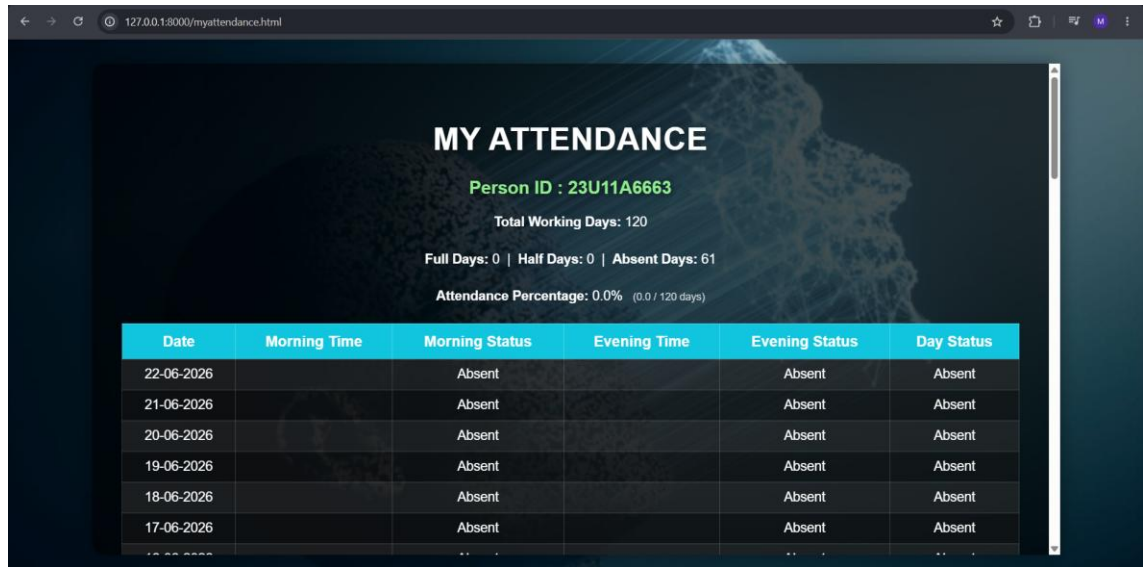


Figure 7.6: Successful Face Recognition and Attendance Marked

The system successfully detects and recognizes a registered user using facial embeddings and CNN-based anti-spoofing verification. Attendance is automatically recorded after successful authentication, and the message "Attendance Marked Successfully" is displayed.

7.1.7 Attendance Record Display



MY ATTENDANCE

Person ID : 23U11A6663

Total Working Days: 120

Full Days: 0 | Half Days: 0 | Absent Days: 61

Attendance Percentage: 0.0% (0.0 / 120 days)

Date	Morning Time	Morning Status	Evening Time	Evening Status	Day Status
22-06-2026		Absent		Absent	Absent
21-06-2026		Absent		Absent	Absent
20-06-2026		Absent		Absent	Absent
19-06-2026		Absent		Absent	Absent
18-06-2026		Absent		Absent	Absent
17-06-2026		Absent		Absent	Absent

Figure 7.7: Attendance Record Display

Attendance details including Person ID, Date, Time, and Status are retrieved from the MySQL database and displayed to the administrator through the View Attendance interface.

7.1.8 Spoof Detection Output

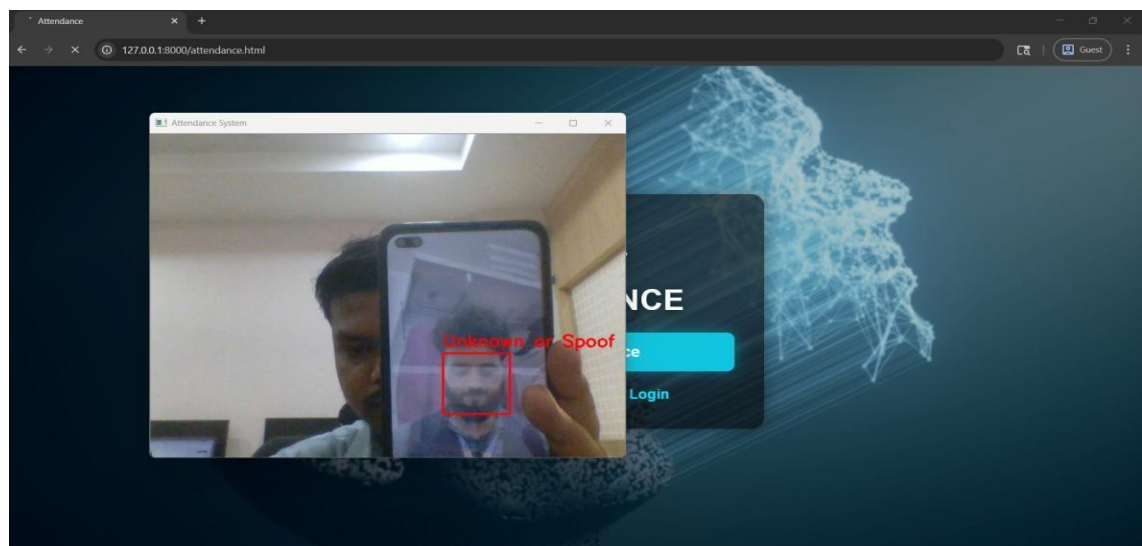


Figure 7.8: Spoof Detection Output

The CNN anti-spoofing model correctly identifies a spoof attempt where the user presents a mobile phone screen displaying a face photograph. The system labels the detection as "Unknown or Spoof" and prevents attendance marking, thereby protecting the system from proxy attendance.

7.1.9 User Deletion

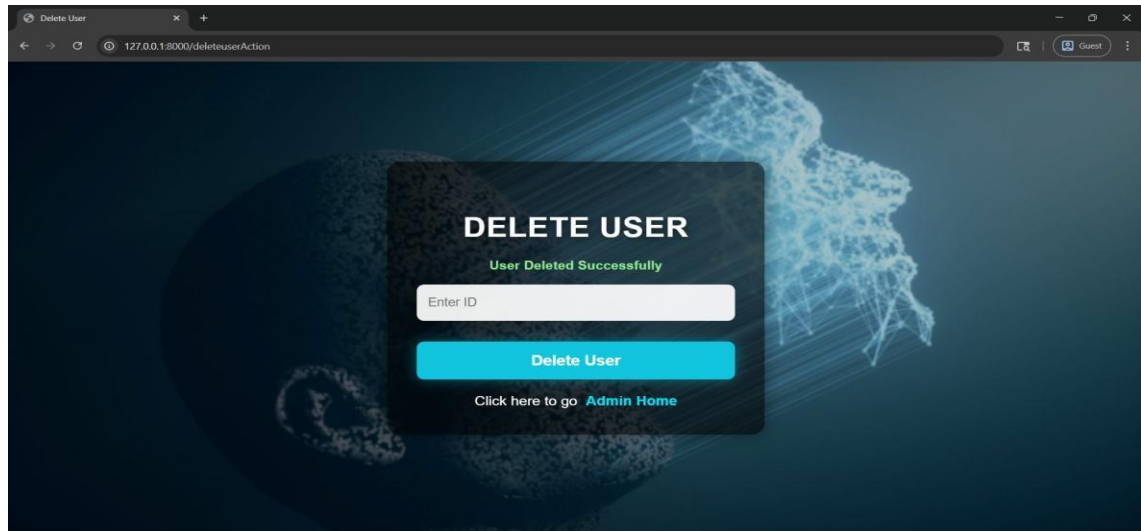


Figure 7.9: User Deletion Interface

The administrator successfully removes a registered user from the system. The User Deleted Successfully message confirms that the user record and associated face encodings have been removed from the database.

7.2 Results Interpretation

The experimental results demonstrate that the proposed Anti-Spoofing Face Attendance System successfully performs face detection, spoof detection, face recognition, and attendance recording. The CNN model effectively distinguishes genuine faces from spoof images, thereby reducing the possibility of proxy attendance.

The face recognition module accurately identifies registered users and records attendance automatically in the MySQL database. Spoof attempts using mobile displays, printed photographs, or unauthorized images are successfully rejected by the system. These results indicate that the integration of anti-spoofing and facial recognition improves both security and reliability of the attendance management process.

The session-based attendance management correctly identifies Morning and Evening attendance sessions. Late attendance is automatically flagged based on predefined time thresholds. Absent records are generated automatically for users who miss attendance sessions, reducing the need for manual intervention

7.3 Performance Evaluation

The performance of the proposed system was evaluated based on recognition accuracy, spoof detection capability, and attendance recording efficiency.

Table 7.1: Performance Evaluation Metrics

Metric	Performance
Face Detection Accuracy	High (Real-time)
Anti-Spoofing Accuracy	97%
Face Recognition Accuracy	High
Attendance Recording	Successful
Response Time	Real-Time (< 3 seconds)
Database Storage	Successful
Late Attendance Detection	Accurate
Absent Generation	Automatic

The system demonstrated stable performance under normal operating conditions and successfully handled genuine and spoof face inputs. Real-time processing was achieved without significant delay.

7.4 Comparative Results

The proposed system was compared with existing attendance systems available in the literature to highlight its advantages.

Table 7.2: Comparative Analysis

System	Face Recognition	Anti-Spoofing	Attendance Analytics	Accuracy
Traditional Attendance	No	No	Manual	N/A
Basic Face Recognition System	Yes	No	Basic	~90%
Blink Detection Based System	Yes	Partial	Basic	92.68%
Proposed System	Yes	Yes (CNN)	Advanced	97%

The comparison shows that the proposed system provides an additional layer of security through CNN-based anti-spoofing detection along with advanced attendance analytics including session management, late detection, absent generation, and attendance percentage calculation. This makes it superior to existing approaches for secure and intelligent attendance management.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

8.1 Summary of Work Done

The Anti-Spoofing Face Attendance System was developed to provide a secure and automated attendance management solution using facial recognition and deep learning techniques. The system integrates face detection, CNN-based anti-spoofing verification, face recognition, session-based attendance management, and MySQL database management to ensure accurate and secure attendance recording.

The developed system successfully captures facial images through a webcam, verifies facial authenticity using a trained CNN model, recognizes registered users using facial encoding comparison, and records attendance automatically. The implementation reduces manual effort, improves attendance accuracy, and prevents unauthorized attendance marking through spoofing attacks.

Advanced features including Morning and Evening session management, Late attendance detection, automatic Absent generation, attendance percentage calculation based on 120 working days, and attendance analytics dashboards have been successfully implemented. Testing results demonstrated that the system performs efficiently in real-time environments and provides reliable attendance management with a spoof detection accuracy of 97%.

8.2 Limitations

Although the proposed system performs effectively, certain limitations exist that should be considered for future improvements.

- The performance of face recognition may be affected by poor lighting conditions.
- Recognition accuracy can decrease when facial images are partially occluded by masks, glasses, or other objects.
- The CNN model requires sufficient training data for improved spoof detection performance.
- System performance depends on the quality of the webcam used for image capture.

- Extreme facial pose variations may affect recognition accuracy.
- The current implementation is primarily designed for small and medium-scale attendance environments.
- The system requires an initial setup and training phase before deployment.
- Network-based spoofing attacks using high-definition 3D masks are not fully addressed in the current implementation.

8.3 Challenges Faced

Several challenges were encountered during the development and implementation of the system that required careful consideration and problem-solving.

- Collecting and preparing a balanced dataset for CNN training with sufficient real and spoof face samples.
- Handling variations in lighting conditions during face detection and recognition.
- Improving anti-spoofing accuracy for different types of spoof attacks including replay videos and screen displays.
- Integrating face detection, spoof detection, face recognition, and database modules into a single cohesive system.
- Managing real-time webcam processing without significant performance delays.
- Ensuring reliable communication between the Django web framework and MySQL database.
- Implementing session-based attendance management with accurate time window detection.
- Reducing false recognition and classification errors during testing.

Despite these challenges, appropriate solutions were implemented to achieve satisfactory system performance and reliability.

8.4 Future Enhancements

The proposed system can be further improved through the following enhancements to increase security, scalability, and usability.

- Integration of advanced deep learning models such as ResNet or Vision Transformers to improve face recognition and anti-spoofing accuracy.
- Support for multiple types of spoof attack detection including 3D mask attacks, deepfake video attacks, and adversarial face attacks.
- Deployment as a cloud-based attendance management platform with multi-institution support.
- Development of a mobile application for remote attendance monitoring and management.
- Integration with institutional management systems and ERP platforms for seamless data exchange.
- Implementation of real-time notification and reporting features through email or SMS alerts.
- Enhancement of system scalability for large organizations and educational institutions with hundreds of users.
- Addition of biometric multi-factor authentication combining face recognition with fingerprint or iris recognition for increased security.
- Integration of geo-location verification to ensure attendance is marked from authorized locations.
- Support for multi-camera surveillance environments to enable attendance marking from multiple entry points simultaneously.

CHAPTER 9

REFERENCES

9.1 Technical Publications (Journals and Conference Papers)

- [1] Arsenovic, M., Sladojevic, S., Anderla, A., and Stefanovic, D., "FaceTime: Deep Learning Based Face Recognition Attendance System", Proceedings of the IEEE International Symposium on Intelligent Systems and Informatics, 2017.

- [2] Mohamed, A., Ahmed, K., and Hassan, M., "Face Anti-Spoofing Using Deep Convolutional Neural Networks", International Journal of Computer Vision and Image Processing, Vol. 11, No. 3, 2021.

- [3] Bagga, A. and Singh, A., "Face Recognition Based Attendance System Using Machine Learning", International Conference on Advances in Computing and Communication Engineering, 2016.

- [4] Ganjoo, M. and Purohit, G., "Automated Attendance System Using Face Recognition and Liveness Detection", International Journal of Engineering Research and Technology (IJERT), Vol. 10, Issue 5, 2021.

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- [8] Simonyan, K. and Zisserman, A., "Very Deep Convolutional Networks for Large-Scale Image Recognition", International Conference on Learning Representations (ICLR), 2015.
- [9] He, K., Zhang, X., Ren, S., and Sun, J., "Deep Residual Learning for Image Recognition", Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016.
- [10] LeCun, Y., Bengio, Y., and Hinton, G., "Deep Learning", Nature, Vol. 521, pp. 436-444, 2015.

9.2 Websites and Online Resources

- [11] OpenCV Documentation – <https://opencv.org>
- [12] TensorFlow Documentation – <https://www.tensorflow.org>
- [13] Keras Documentation – <https://keras.io>
- [14] MySQL Documentation – <https://dev.mysql.com/doc>
- [15] Python Documentation – <https://docs.python.org/3>
- [16] Face Recognition Library Documentation – <https://face-recognition.readthedocs.io>
- [17] Django Documentation – <https://docs.djangoproject.com>
- [18] Stack Overflow Developer Community – <https://stackoverflow.com>
- [19] GitHub Repository – <https://github.com/manideepreddy-2004/Anti-Spoofing-Face-Attendance-System>

9.3 Other Sources

[20] Project source code developed using Python, TensorFlow, OpenCV, Django, and MySQL.

[21] Dataset images collected and processed for CNN training and testing purposes.

[22] Mini Project Documentation Guidelines provided by the Department of Computer Science and Engineering (AIML), Samskruti College of Engineering and Technology, 2025-2026.

CHAPTER 10

APPENDICES

Appendix A: Software Development Life Cycle (SDLC) Documents

A.1 Software Requirement Specification (SRS)

The complete Software Requirements Specification is provided in Chapter 3 of this document, covering functional requirements, non-functional requirements, hardware and software specifications, and user characteristics.

A.2 Feasibility Study

The feasibility study covering Technical Feasibility, Economic Feasibility, and Operational Feasibility is detailed in Section 3.1 of this document.

A.3 Test Reports

- Unit Testing Report: Refer to Section 6.2
- Integration Testing Report: Refer to Section 6.3
- System Testing Report: Refer to Section 6.4

Appendix B: Project Timeline

Table B.1: Project Schedule (Gantt Chart)

Activity	Week 1	Week 2	Week 3-4	Week 5-7	Week 8	Week 9
Requirement Analysis	Done					
Literature Survey		Done				
System Design			Done			
Implementation				Done		
Testing					Done	
Documentation						Done

Appendix C: Cost Estimation

Table C.1: Cost Estimation Sheet

Resource	Estimated Cost
Python Software	Free (Open Source)
OpenCV Library	Free (Open Source)
TensorFlow Framework	Free (Open Source)
Keras Library	Free (Open Source)
Django Web Framework	Free (Open Source)
MySQL Database	Free (Open Source)
Development System	Existing Resource
Webcam	Existing Device
Total Cost	Minimal

Appendix D: Ethical Considerations and Consent

The Anti-Spoofing Face Attendance System is developed for educational and research purposes under the guidance of the Department of Computer Science and Engineering (AIML), Samskruti College of Engineering and Technology.

The following ethical considerations were followed during the development:

- Facial images are collected only with explicit user consent.
- Collected facial data is used exclusively for attendance verification and identity authentication.
- User information and attendance records are stored securely in the database.
- Access to facial data and attendance records is restricted to authorized administrators only.
- No user data is shared with third parties or used for commercial purposes.
- Users have the right to request deletion of their data from the system.

Appendix E: Plagiarism Report

Not Applicable.

Appendix F: Source Code Repository and Deployment Links

The complete source code of the Anti-Spoofing Face Attendance System is available at the following GitHub repository:

GitHub Repository: <https://github.com/manideepreddy-2004/Anti-Spoofing-Face-Attendance-System>

The repository contains:

- Complete Python source code
- Django web application files
- CNN model training scripts
- Face detection and recognition modules
- Database schema and setup scripts
- Requirements file for dependency installation

Appendix G: Additional Screenshots and User Manual

Additional screenshots related to the following system functionalities are provided in this section.

G.1 CNN Training Output

```
Epoch 1/10 [=====] - 9s 98ms/step - loss: 0.6920 - accuracy: 0.5228 - val_loss: 0.6643 - val_accuracy: 0.6023
Epoch 2/10 [=====] - 6s 83ms/step - loss: 0.6077 - accuracy: 0.6835 - val_loss: 0.5451 - val_accuracy: 0.7414
Epoch 3/10 [=====] - 6s 83ms/step - loss: 0.4926 - accuracy: 0.7691 - val_loss: 0.4248 - val_accuracy: 0.8145
Epoch 4/10 [=====] - 6s 83ms/step - loss: 0.3802 - accuracy: 0.8365 - val_loss: 0.3307 - val_accuracy: 0.8698
Epoch 5/10 [=====] - 6s 83ms/step - loss: 0.2901 - accuracy: 0.8902 - val_loss: 0.2624 - val_accuracy: 0.9023
Epoch 6/10 [=====] - 6s 83ms/step - loss: 0.2083 - accuracy: 0.9264 - val_loss: 0.2080 - val_accuracy: 0.9261
Epoch 7/10 [=====] - 6s 83ms/step - loss: 0.1513 - accuracy: 0.9498 - val_loss: 0.1761 - val_accuracy: 0.9382
Epoch 8/10 [=====] - 6s 83ms/step - loss: 0.1056 - accuracy: 0.9656 - val_loss: 0.1587 - val_accuracy: 0.9444
Epoch 9/10 [=====] - 6s 83ms/step - loss: 0.0781 - accuracy: 0.9764 - val_loss: 0.1419 - val_accuracy: 0.9508
Epoch 10/10 [=====] - 6s 84ms/step - loss: 0.0551 - accuracy: 0.9842 - val_loss: 0.1312 - val_accuracy: 0.9556
===== RESULT =====
Training Accuracy : 0.9842
Validation Accuracy : 0.9556
Spoof Model Trained Successfully
Model Saved Inside models/spoof_model.h5
History Saved Inside models/cnn_history.pkl
```

Figure G.1(a): CNN Training Process Output (Console)

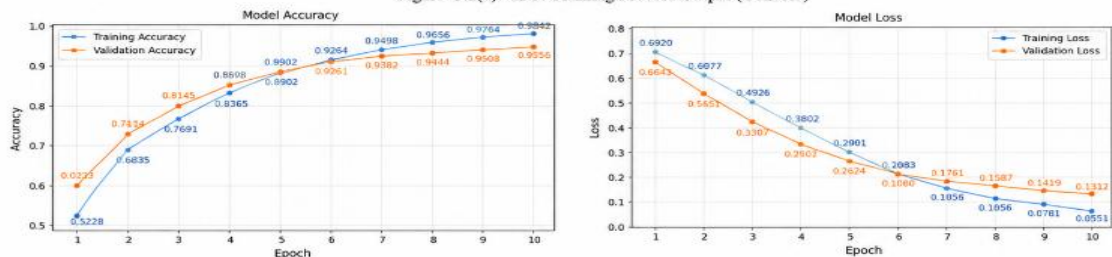


Figure G.1: CNN Training Accuracy and Loss Graph

G.2 Attendance Analytics Dashboard

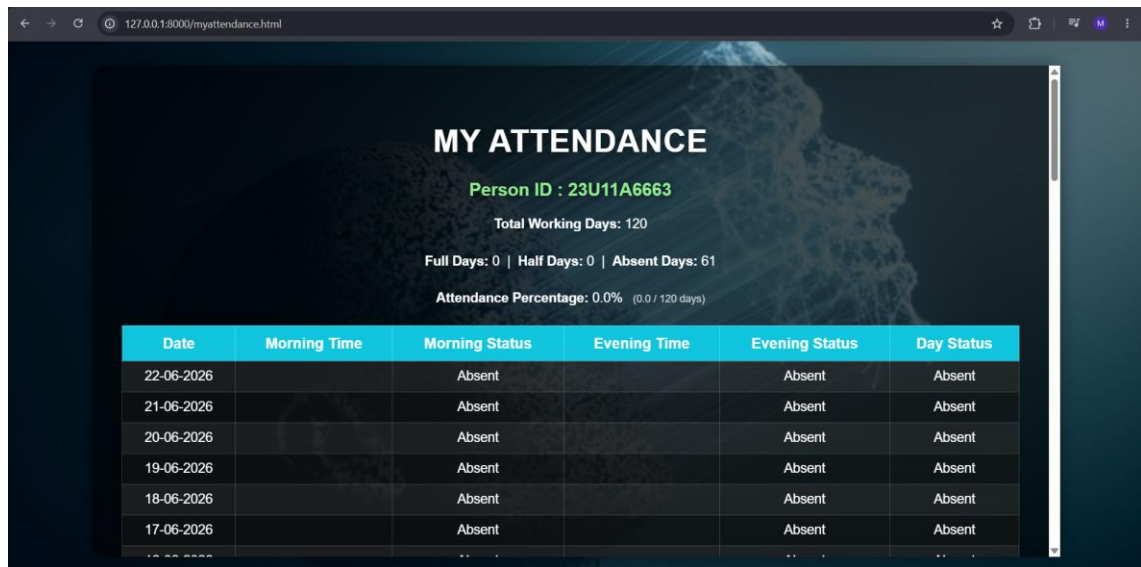


Figure G.2: Attendance Analytics Dashboard

G.3 User Manual Summary

Administrator Steps:

10. Login using admin credentials on the Admin Login page.
11. Register new users using the Register User interface by providing Person ID and Name.
12. Ensure the user faces the webcam clearly during registration.
13. Upload face dataset and train the CNN model if required.
14. View attendance records by entering the Person ID in the View Attendance interface.
15. Delete users from the system using the Delete User interface.

User Steps:

16. Navigate to the attendance page and enter your Person ID.
17. Click Start Attendance to activate the webcam.
18. Face the webcam clearly without accessories obstructing the face.
19. Wait for the system to recognize your face and confirm attendance.
20. Attendance is marked automatically upon successful authentication.

Appendix H: Journal / Conference Paper Publication

No journal or conference paper has been published as part of this project. The work is intended for academic submission as a Mini Project Report for the B.Tech degree in Computer Science and Engineering (AIML) at Samskruti College of Engineering and Technology.

Appendix I: Participation Certificates

Not Applicable for this submission.

Appendix J: Project Submission DVD Contents

The DVD submitted along with this project report contains the following materials:

21. Project Source Code – Complete Python and Django source code
22. Project Documentation – Soft copy of this project report in PDF and Word format
23. Review Presentation Slides – PPT files for all project reviews (Review 1, Review 2, and Final Review)
24. Demonstration Video – A screen-recorded video demonstrating the complete project flow and execution
25. Dataset Files – Face image dataset used for CNN training and testing
26. Database Backup – MySQL database backup file (faceattendance.sql)
27. CNN Model File – Trained CNN model file used for anti-spoofing detection
28. README File – Instructions for setting up and running the project